DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY
UTTAR PRADES, LUCKNOW



Evaluation Scheme & Syllabus For

B.Tech. 3rd Year (Food Technology)

Based on

AICTE MODEL CURRICULUM & NEP2020

(Effective from the Session: 2024-25)

B.TECH (Food Technology)

SEMESTER-V												SESSION2024-25	
Sl. No	Subject Codes	Subject	Periods			Evaluation Scheme				End Semester		Total	Credit
			L	T	P	CT	TA	Total	PS	TE	PE		
1	BCH 504	CHEMICAL REACTION ENGINEERING	3	1	0	20	10	30		70		100	4
2	BCH 505	CHEMICAL ENGINEERING THERMODYNAMICS	3	1	0	20	10	30		70		100	4
3	BFT501	PRINCIPLES OF FOOD PRESERVATION	3	1	0	20	10	30		70		100	4
4	BFT 051-053	Departmental Elective-I	3	0	0	20	10	30		70		100	3
5	BFT 054-056	Departmental Elective-II	3	0	0	20	10	30		70		100	3
6	BCH554	CRE LAB	0	0	2				50		50	100	1
7	BFT551	FOOD PRESERVATION LAB	0	0	2				50		50	100	1
8	BFT552	FOOD PRODUCTSLAB-I	0	0	2				50		50	100	1
9		Mini Project or Internship Assessment*	0	0	2				100			100	1
10	BNC501/ BNC502	Constitution of India/ Essence of Indian Traditional Knowledge	2	0	0	20	10	30		70			NC
11		MOOCs(Essential for Hons .Degree)											
		Total	**17**	**3**	**8**							**900**	**22**

*The Mini Projector internship (4weeks) conducted during summer break after IV semester and will be assessed during V semester.

SEMESTER-VI												SESSION2024-25	
Sl. No	Subject Codes	Subject	Periods			Evaluation Scheme				End Semester		Total	Credit
			L	T	P	CT	TA	Total	PS	TE	PE		
1	BCH 604	INSTRUMENTATION & PROCESS CONTROL	3	1	0	20	10	30		70		100	4
2	BFT601	BAKERY TECHNOLOGY	3	1	0	20	10	30		70		100	4
3	BFT602	DAIRY TECHNOLOGY	3	1	0	20	10	30		70		100	4
4	BFT061-063	Departmental Elective-III	3	0	0	20	10	30		70		100	3
5		Open Elective-I	3	0	0	20	10	30		70		100	3
6	BCH 654	INSTRUMENTATION & PROCESS CONTROL LAB	0	0	2				50		50	100	1
7	BFT651	BAKERY TECHNOLOGY LAB	0	0	2				50		50	100	1
8	BFT652	DAIRY TECHNOLOGY LAB	0	0	2				50		50	100	1
9	BNC601/ BNC602	Constitution of India/ Essence of Indian Traditional Knowledge	2	0	0	20	10	30		70			
10		MOOCs(Essential for Hons. Degree)											
		Total	**17**	**3**	**6**							**800**	**21**

DEPARTMENTELECTIVE–1

BFT-051	CEREALS, PULSES & OILSEED PRODUCTS
BFT-052	ENGINEERED, FABRICATED & TEXTURED FOODS
BFT-053	FOOD PROCESS ENGINEERING

DEPARTMENTELECTIVE–2

BFT-054	FRUITS, VEGETABLES & PLANTATION PRODUCTS
BFT-055	NUTRACEUTICALS & FUNCTIONA L FOODS
BFT-056	TECHNOLOGY OF FOOD BEVERAGES

DEPARTMENTELECTIVE-3

BFT-061	TECHNOLOGY OF ANIMAL FOODS
BFT-062	TRADITIONAL & FERMENTED FOODS
BFT-063	SPECIALITY FATS FOR FOOD INDUSTRY

DR.A.P.J.ABDUL KALAM TECHNICAL UNIVERSITY

B.TECH III YEAR V SEMESTER FOOD TECHNOLOGY COURSE

SUBJECT CODE: BCH504	COURSE TITLE: CHEMICAL REACTION ENGINEERING
EXAM DURATION: 03 HOURS	SEMESTER: V
L: T: P::3 :1 : 0 CREDITS: 4	PRE REQUISITES: NIL

OBJECTIVES:

- To impart the basic concepts of Chemical Reaction Engineering
- To develop understanding about reactor analysis, performance and design.
- To develop understanding of Reaction Kinetics

COURSEOUTCOME:

1. Understand the factors influencing reaction rates and analyze reaction mechanisms.
2. Interpret kinetic data using integral and differential methods for various reaction orders.
3. Understand about reactor analysis and design.
4. Decide the type of reactors and their arrangement of reactors for best performance.
5. Analyse the effect of temperature and pressure on the reaction kinetics
6. Analyze non-ideal flow patterns using residence time distribution and related curves.

CO/PO	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	3	3	2	1	1	-	-	-	-	-	-	-
CO-2	3	3	2	3	2	-	-	-	-	-	-	-
CO-3	3	2	3	2	3	-	-	-	-	-	-	-
CO-4	3	3	3	2	3	-	-	-	-	-	-	-
CO-5	3	2	2	3	2	-	-	-	-	-	-	-
CO-6	3	2	2	2	3	-	-	-	-	-	-	-
Avg	**3.0**	**2.5**	**2.3**	**2.2**	**2.3**	**-**	**-**	**-**	**-**	**-**	**-**	**-**

REFERENCEBOOKS

S.No	NAMEOFAUTHORS/BOOKS/PUBLISHERS	YEAR OF PUBLICATION /REPRINT
1	Smith,J,M,"ChemicalEngineeringKinetics",3rdEdition,McGraw-Hill (1990).	1990
2	Levenspiel,O.,"ChemicalReactionEngineering",3rdEdition,JohnWiley,(1998).	1998
3	FoglerH.S.,ElementsofChemicalReactionEngineering,4thedition,Prentice Hall of India,(2008)	1997

COURSEDETAILS

UNITS	CONTENTS	LECTURE HOURS
I	Chemical Reactions: Rate of chemical reactions, variable affecting the reaction rate, order of reaction, rate constant, elementary and non-elementary reaction mechanism. Temperature dependency from thermodynamics, Arrhenius collision and activated complex theories.	**8**
II	Integral and differential methods for analyzing kinetic data, Interpretation of constant volume batch reactor data for zero, first, second and third order reactions, Half life period, Irreversible reaction in parallel and series, Autocatalytic reaction.	**8**

III	Interpretation of reactor data in variable volume batch reactions, Design equations forbatch, plug flow, back mix flow and semi batch reactors for is o thermal, adiabatic homogeneous reaction.	**8**
IV	Space velocity and residence time in flow reactors, Design of batch, plug flow and mixed flow reactors for first and second order single reactions, Size comparison of single reactorslikebatch,plugflowandCSTRforfirstandsecondordersinglereactions.Multiplereactor systems, Plug flow reactions in series and for parallel equal sized CSTR's in series. Best arrangement of different type of reactors.	**9**
V	Temperatureandpressureeffectsforsinglereaction,Optimaltemperatureprogressionforfirstorder reactions.IntroductiontoNon-Idealflow,Residencetimedistributionoffluidin vessels, E ,F and C curves &their relation.	**9**
	TOTAL	**42**

SUBJECTCODE:BCH505	COURSETITLE:CHEMICAL ENGINEERING THERMODYNAMICS
EXAMDURATION:3 Hours	SEMESTER:V
L: T: P::3:1 : 0 CREDITS: 4	PREREQUISITES: NIL

OBJECTIVES:

- To understand the laws of Thermodynamics in solving problems
- To impart fundamental concepts of solution Thermodynamics to describe& predict phase equilibria
- To understand the functional relationship between Chemical Reactions and Thermodynamics

COURSEOUTCOME:

1. Understand and apply the fundamental laws of thermodynamics to engineering problems.
2. Analyze vapor-liquid equilibria using Raoult's law, Henry's law, and modified Raoult's law.
3. Evaluate solution thermodynamics, including chemical potential and partial molar properties, in multi-component systems.
4. Apply thermodynamic models to predict phase equilibria and assess excess Gibbs energy in mixtures.
5. Determine equilibrium constants and analyze the impact of temperature on chemical reaction equilibria.
6. Perform thermodynamic analysis of single and simultaneous reactions to evaluate equilibrium conversion.

CO/PO	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	3	2	2	1	2	-	-	-	-	-	2	-
CO-2	3	3	2	2	2	-	-	-	-	-	2	-
CO-3	3	3	2	2	2	-	-	-	-	-	2	-
CO-4	3	3	2	2	3	-	-	-	-	-	2	-
CO-5	3	3	2	2	2	-	-	-	-	-	2	-
CO-6	3	3	2	2	2	-	-	-	-	-	2	-
Avg	**3.0**	**2.8**	**2.0**	**1.8**	**2.2**	**-**	**-**	**-**	**-**	**-**	**2**	**-**

REFERENCEBOOKS

S.O	NAMEOFAUTHORS/BOOKS/PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1	Smith,J.M.,VanNess,H.C.&Abbot,M.M."IntrotoChemicalEngineeringThermodynamics", 5th edition. NewYork: Mc-Graw Hill	1996
2	DaubertT.E., "Chemical Engineering Thermodynamics" Mc Graw Hill	1986
3	Y.V.C.Rao,"Chemical Engineering Thermodynamics" University press	1997

COURSEDETAILS

UNIT	CONTENTS	LECTURE HOURS
I	Basic concepts in Thermodynamics, laws of Thermodynamics and their applications in engineering problems, steady flow processes, unsteady flow processes, entropy and entropy balance for open systems	**9**
II	An introduction to vapor-liquid Equilibria, qualitative behavior of the vapour-liquid equilibria (VLE), simple models for vapor-liquid equilibria, Raoults law, Henry's law, Dew point and Bubble point calculations, VLE by modified Raoults law, the gamma/Philips formulation of VLE	**8**

III	Solution Thermodynamics- Fundamental property relay, Chemical potential, partial molar Property, partial properties in binary solutions, equations relating molar and partial molar properties, criteria for Equilibrium between phases in multicomponent non reacting systems in terms of chemical potential, fugacity and fugacity coefficient for pure species, fugacity and fugacity coefficient for species in solutions, fugacity coefficient from the virial equation of state and generalized correlations	**9**
IV	Solution Thermodynamics Applications- ideal solution, Lewis rand all rule, excess Gibbs energy and activity coefficient, Liquid phase properties from VLE data, Excess Gibbs energy data reduction, Thermodynamic consistency of phase equilibria, applications of the correlations and prediction of phase equilibria, models for excess Gibbs energy, Property change of mixing	**8**
V	Chemical Reaction Equilibria-The reaction coordinate, standard Gibbs free energy change and Equilibrium constant, Effect of temperature on Equilibrium constant, evaluation of Equilibrium constant, relation of equilibrium constants to composition for gas phase and liquid phase reactions, Equilibrium conversion for single reaction, Thermodynamic analysis of simultaneous reactions	**8**
	TOTAL	**42**

SUBJECT CODE: KFT-501	COURSE TITLE:PRINCIPLES OF FOOD PRESERVATION
EXAM DURATION:	SEMESTER:V
L: T: P:: 3 : 1: 0 CREDITS :4	PRE REQUISITES: NIL

OBJECTIVES:

- **To provide basic knowledge of Principles of Food Preservation**
- **To provide in depth knowledge of Technologies based on these principles**
- **To study the working principles, Advantages and Limitations of these processes**

COURSEOUTCOME:

After successful completion of the course the students will be able to:

1. Understand the reasons/causes of food spoilage and the need for Food Preservation
2. Understand the basic principles of Food Preservation & technologies available for the same.
3. Understand the equipments associated with these processes and their working principle
4. Understand the suitable use of these technologies for practical purposes
5. Understand the impact of these process technologies on product and environment
6. Understand the latest developments in related process technologies

COs	Bloom's Level	POs 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) "-": No Correlation											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	K3	3	1	-	1	-	1	1	-	-	-	-	1
CO2	K2	3	2	-	1	-	1	-	-	-	-	-	1
CO3	K2	3	2	1	1	-	1	-	-	-	-	-	1
CO4	K3	3	2	1	1	-	1	-	-	-	-	-	1
CO5	K3	3	2	2	1	-	3	2	-	-	-	-	1
CO6	K3	3	2	3	1	-	2	-	-	-	-	-	1
Average		3	1.8	1.2	1	-	1.5	0.5	-	-	-	-	1

REFERENCEBOOKS

S.NO	NAMEOFAUTHORS/BOOKS/PUBLISHERS	YEAR OF PUBLICATION /REPRINT
1	Norman N.Potter, Joseph H.Hotchkiss,"FoodScience",5thEdition,CBS Publishers & Distributor Pvt. Ltd.	2007
2	Marcus Karel,Owen R. Fennema, Daryl B.Lund, "Principles of Food Science, Part-II, Physical Principles of Food Preservation, M.Dekker	1975, Digitized April2008
3	Norman W.Desrosier, Technology of Food Preservation, Medtech, Scientific International Pvt. Ltd New Delhi	2018
4	C.R.Stumbo," Thermobacteriology in FoodProcessing"2nd Edition,Academic Press	1973

Course details

UNITS	S.NO	CONTENTS	LECTURE HOURS
I	1	Basic considerations: Aims and objectives of preservation & processing of foods, Wastage of Foods, Food Safety and Food Security, Causes of quality deterioration and spoilage of perishable foods	4

	2	Characteristics of tissue and non-tissue foods, Degree of perishability of unmodified foods, Concept of Intermediate moisture foods, Basic Principles of Food Preservation and Methods Involved	**4**
II	**1**	Preservation of foods by low temperatures: Chilling temperatures: Consideration relating to storage of foods at chilling temperatures, Applications and procedures, Controlled and Modified atmosphere storage of foods, Post storage Handling of foods. Advantages and Limitations of Chilling Process	**4**
	2	Freezing temperatures: Freezing process, Slow and fast freezing of foods and its consequence, other occurrences associated with freezing of foods. Technological aspects of Pre freezing, Actual freezing, Frozen storage and Thawing of foods. Technologies involved and their working principles.	**5**
III	**1**	Preservation of foods by high temperatures: Basic concepts in thermal destruction of microorganisms. D, Z, F values. Heat resistance and thermophic microorganisms. Thermal Death Time Curve and its implications. Cooking, Blanching, Pasteurization and Sterilization of foods.	**5**
	2	General process of Caning of foods, Spoilage in Canned foods. Canning in Retortable Pouches. Assessing adequacy of thermal processing of foods.	**5**
IV	**1**	Preservation of Foods by water removal: Basic Principles, Technological aspects and application of evaporative concentration process; Freeze concentration and membrane process for food concentrations and their applications along with advantages and limitations.	**5**
	2	Basic Principles, Technological aspects and application of drying and dehydration of foods, Cabinet, tunnel, belt, bin, drum, spray, vacuum, foam mat, fluidized-bed and freeze drying of foods. Basic Principles, working principle and equipments involved. Advantages and Limitations of these processes.	**5**
V	**1**	Principles, Technological aspects and application of sugar and salt in food preservation. Use of Preservatives in Food Preservation, Class I and Class II Preservatives, Antimicrobial agents, Biological agents. Basic Principles and limits of usage as per FSSAI.	**4**
	2	Use of Non ionizing and Ionizing radiations in preservation of foods. Basic Principles. Equipments involved and working principles. Advantages and Limitations. Safety aspects of Irradiated Foods. Hurdle technology and its application in production of Shelf Stable Foods.	**4**

SUBJECT CODE: KFT-051	COURSE TITLE: CEREALS, PULSES & OILSEED PRODUCTS
EXAMINATION DURATION: 3Hrs.	SEMESTER: V
L:T:P: :3: 0: 0 CREDITS: 3	PRE-REQUISITE: NIL

OBJECTIVES:

- **To provide basic knowledge of storage, losses, nutritional value, milling and processing of cereal grains, pulses & oilseeds.**
- **To provide knowledge of Milling processes and equipments involved**
- **To provide knowledge of Quality Aspects of milled products**

COURSE OUTCOME:

After successful completion of the course the students will be able to:

1. Understand the postharvest losses, storage and nutritional value of grains.
2. Understand the characterizion of grains, pulses and oilseeds on the basis of milling characteristics.
3. Understand milling and subsequent processing of milled products for value addition.
4. Understand the quality aspects of milled products and the relevant parameters and their estimation.
5. Understand the processing technologies adopted for oilseeds and edible oil extraction
6. Understand the development of value added and tailor made products from plant sources.

COs	Bloom's Level	POs 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) "-": No Correlation											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	K3	2	1	1	-	-	-	-	-	-	-	-	-
CO2	K2	2	1	1	-	-	-	-	-	-	-	-	1
CO3	K2	1	1	3	-	2	-	-	-	-	-	-	1
CO4	K3	1	1	1	-	-	-	-	-	-	-	-	1
CO5	K3	3	2	3	-	3	2	-	-	-	-	-	1
CO6	K3	3	1	3	-	3	3	-	-	-	-	-	1
Average		2	1.2	2.0	-	1.3	0.9	-	-	-	-	-	0.9

REFERENCE BOOKS:

S.No	Name of Authors / Books / Publishers	Year of Publication /Reprint
1.	Chakravarti A.: Postharvest Technology of Cereals, Pulses and Oilseeds, Oxford andIBH, New Delhi	2019
2.	Y.Pomeranz: Cereals and Cereal Products, Wiley online library https://doi.org/10.1002/14356007.a06_093	2000
3.	S. A. Matz: Chemistry and Technology of Cereals as Food and Feed, CBS Publishers &Distributors	2018
4.	S. A. Matz. : Cereal Technology. , Ed. AVI Publishing Co., Westport	

COURSEDETAILS:

UNITS	S.NO.	CONTENTS	LECTURE HOURS
I	1	Composition, Structure and Processing characteristic of Cereal grains, Legumes and oilseeds. Post harvest, Post processing practices for their safe storage.	4

	2	Parboiling and Milling of paddy, Quality characteristics, Curing and aging of rice, Processed rice products.	**4**
II	**1**	Wheat and its quality characteristics. International and Domestic processes adopted for milling Wheat into flour and semolina, Turbo grinding of Wheat and air classification: Objectives and Processes adopted along with equipment involved.	**4**
	2	Flour grades and their suitability for baking purposes, Assessment of flour quality and characteristics Milling of Durum wheat, Macaroni products	**4**
III	**1**	Principles and Technologies involved in Dry and Wet milling of corn	**4**
	2	Starches and their conversion products, Cornflakes Manufacture Malting of barley: Objectives and processes adopted	**4**
IV	**1**	Milling of legume-pulses by traditional and improved processes.	**4**
	2	Purpose, Principles and Technologies involved in Pearling of Millets	**3**
V	**1**	Processing of oil seeds for direct use and consumption, Oil and protein products.	**2**
	2	Processing of extracted oil. Refining, Hydrogenation, Interestrification. Peanut butter, Introduction to Margarine and Spread.	**7**
	3	Processing of deoiled cake into protein concentrates and isolates, Textured protein, Functional protein preparations.	**3**

SUBJECT CODE: KFT-052	COURSE TITLE: ENGINEERED, FABRICATED AND TEXTURED FOODS
EXAM DURATION: 3.0 HOURS	SEMESTER : V
L: T: P ::3 : 0 : 0 CREDITS: 3	PRE REQUISITES: NIL

OBJECTIVES:

- **To provide knowledge of special types of foods that are engineered, modified/ fabricated to suit some special purpose or need as per the customer / food industry requirements**
- **To provide basic knowledge of raw materials, processes and equipments involved.**
- **To gain insight into the quality characteristics of such food products.**

COURSE OUTCOME:

After successful completion of the course the students will be able to:

1. Understand the requirements of Engineered, Fabricated and Textured Foods
2. Understand the basic principles and functional importance of these foods
3. Understand the technology of fabrication and its application in different foods.
4. Understand the equipments associated with the processes and their working principle
5. Understand the technology of extrusion and various types of extruders
6. Understand the recent technologies and advancements in the development of these products.

COs	Bloom's Level	POs 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) "-": No Correlation											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	K3	3	3	1	1	2	2	-	-	-	-	-	1
CO2	K2	3	1	1	1	2	-	-	-	-	-	-	1
CO3	K2	3	1	1	1	2	2	-	-	-	-	-	1
CO4	K3	3	2	2	1	3	-	-	-	-	-	-	1
CO5	K3	3	2	2	2	3	-	-	-	-	-	-	1
CO6	K3	3	2	3	2	3	2	-	-	-	-	-	1
Average		3.0	1.8	1.7	1.3	2.5	1.0	-	-	-	-	-	1.0

REFERENCEBOOKS

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1	Norman N.Potter, Joseph H.Hotchkiss, " Food Science", 5th Edition, CBS Publishers & Distributors Pvt. Ltd.	2007
2	Zeki Berk," Food Process Engineering & Technology", 3rd Edition, Academic Press	2018
3	Weibiao Zhou , Y. H. Hui ,I. De Leyn ,M. A. Pagani ,C. M. Rosell , J. D. Selman , N.Therdthai, "Bakery Products Science and Technology", Second Edition, Wiley Online	2014
4	Shubhangini A. Joshi, "Nutrition and Dietetics" 5th Edition, Mc Graw Hill India	2021
5	S.A.Matz, The Chemistry and Technology of Cereals as Food and Feed, Kindle Edition, CBS Publishers and Distributors Pvt. Ltd, New Delhi	2018

COURSE DETAILS

UNITS	S.NO	CONTENTS	LECTURE HOURS
I	1	Extrusion Process, Principles of Extrusion and Extruders. Single Screw & Multiple Screw Extruders. Working Principle and Operations.	5

	2	Physico Chemical Changes in Foods during Extrusion process. Extrusion cooking. Physical & Chemical Changes in foods during Extrusion Cooking. Puffing Operations, Puffing Gun and Puffed Food	**5**
II	**1**	Textured vegetable protein products., Soy Proteins, Soy Milk, Soy Paneer, Protein Concentrates and Protein Isolates, Puffed Products. Meat Analogues., Imitation Paneer .	**8**
III	**1**	Fabricated RTS Beverages, Raw Materials and their Characteristics, Industrial Manufacture of RTS Beverages	**4**
	2	Industrial Production of Special Bakery Products, Margarine, Peanut Butter, Imitation Milks, Non Dairy based Ice Creams, Designer Lipids etc. Raw Materials, Equipment Involved and Product Characteristics.	**4**
IV	**1**	Definition of Weaning Foods. Formulation of Weaning Foods. Raw Materials, their characteristics and role.	**4**
	2	Concept, Definition and Formulation of Baby Foods. Therapeutic Foods . Geriatric Foods. Raw Materials and their role. Manufacturing Process, Product Characteristics.	**4**
V	**1**	Macaroni, Pasta, Noodles, Vermicelli etc. Raw Materials used and their industrial manufacture. Product Characteristics.	**4**
	2	Recent advances, new developments and products in the field of engineered, fabricated and textured foods.	**4**

SUBJECT CODE: KFT-053	COURSE TITLE: FOOD PROCESS ENGINEERING
EXAMINATION DURATION: 3Hrs.	SEMESTER: V
L: T: P: : 3: 0: 0 CREDITS: 3	PRE-REQUISITE: NIL

OBJECTIVES:

- **To provide basic knowledge of engineering and design concepts relevant to food processing industries.**
- **To understand the practical use of these concepts as applicable in Food Processing Industries**

COURSE OUTCOME:

After successful completion of the course the students will be able to:

1. Understand different unit operations applied in food processing industry.
2. Understand the kinetics of various chemical and biochemical reactions in foods.
3. Understand the various Heat and Mass transfer calculations pertaining to various food processes.
4. Understand the rheological behavior of food materials
5. Understand the industrial Refrigeration & Freezing concepts and principles of foods
6. Understand the design considerations and layout of food processing units.

COs	Bloom's Level	POs 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) "-": No Correlation											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	K3	3	3	2	1	2	3	-	-	-	-	-	-
CO2	K2	3	3	2	1	1	2	-	-	-	-	-	-
CO3	K2	3	2	2	2	2	3	-	-	-	-	-	-
CO4	K3	3	2	3	2	2	2	-	-	-	-	-	2
CO5	K4	3	2	3	2	2	3	-	-	-	-	-	2
CO6	K4	3	3	2	2	2	2	-	-	-	-	-	2
Average		3.0	2.5	2.3	1.7	1.8	2.5	-	-	-	-	-	1.0

REFERENCEBOOKS:

S.No	Name of Authors / Books / Publishers	Year of Publication/ Reprint
1.	Albert Ibarz, Gustavo V. Barbosa-Canovas, Introduction to Food Process Engineering, CRC Press	2014
2.	Smith P.G., Introduction to Food Process Engineering, 2nd Edition (Food Science Text Series) Springer	2011
3.	Romeo T. Toledo, Fundamentals of Food Process Engineering, 4th Edition, Springer	2018

COURSEDETAILS:

UNITS	S.NO.	CONTENTS	LECTURE HOURS
I	1	Introduction , Concept of Unit operation, Units and dimensions, Unit conversions, dimensional analysis	3
	2	Mass and Energy Balances in various food processing operations. Grinding and mixing principle and equipments used in food industry. Related numericals	5

II	**1**	Fluid Flow in food Processing, Liquid Transport systems, Properties of Liquids. Newton's Law of Viscosity , Principle of Capillary tube and rotational viscometer ,Properties of Newtonian and Non-Newtonian fluids	**5**
	2	Flow characteristics, Reynolds Number, Bernoulli's Equation , Concept of Flow Measurement devices, Related basic numerical	**4**
III	**1**	Calculations pertaining to Heat Transfer and Flow properties of liquid foods in Transportation and Processing.	**4**
	2	Calculations pertaining to Kinetics of Chemical and Biochemical Reactions in Foods. Thermalstabilization of foods.	**4**
IV	**1**	Refrigeration and Freezing ,Concept and selection of a refrigerant , Freezing time calculations	**4**
	2	Frozen food storage , Refrigeration requirements during storage, Related basic numerical and Refrigeration Load calculations	**4**
V	**1**	Membrane processes in the food industry. Membrane Systems, Membrane Materials Membrane Configurations.	**4**
	2	Calculations pertaining to Mass transfer through MF and UF membranes. Mass transfer in reverse osmosis	**4**

SUBJECT CODE: KFT-054	COURSE TITLE: FRUITS, VEGETABLES AND PLANTATION PRODUCTS
EXAM DURATION: 3.0 HOURS	SEMESTER : V
L: T: P :: 3 : 0 : 0 CREDITS: 3	PRE REQUISITES: NIL

OBJECTIVES:

- **To provide basic knowledge of the physico-chemical composition of fruits and vegetables**
- **To provide knowledge regarding the technology for manufacture of value added products from them.**
- **To provide knowledge about defects, causes and their prevention**
- **To provide knowledge about quality parameters of these value added products**

COURSE OUTCOME:

After successful completion of the course the students will be able to:

1. Understand the composition and characteristics of fresh produce
2. Understand the causes of post harvest losses and basic principles of preservation.
3. Understand the manufacture of Value Added products from excess fresh produce
4. Understand the causative factors of spoilage and means to prevent this spoilage
5. Understand the processing technologies involved in production of minor and major Indian spices.
6. Understand the processing technologies of main plantation products as Tea, Coffee and Cocoa etc.

COs	Bloom's Level	POs 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) "-": No Correlation											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	K3	2	2	2	1	-	-	-	-	-	-	-	-
CO2	K2	3	2	2	2	-	-	-	-	-	-	-	-
CO3	K2	3	2	2	2	-	-	-	-	-	-	-	-
CO4	K3	3	2	2	2	-	-	-	-	-	-	-	-
CO5	K2	3	2	2	2	-	-	-	-	-	-	-	1
CO6	K2	3	2	2	2	-	-	-	-	-	-	-	1
Average		2.8	2.0	2.0	1.8	-	-	-	-	-	-	-	0.3

REFERENCEBOOKS

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION /REPRINT
1	Norman N.Potter, Joseph H.Hotchkiss, " Food Science", 5th Edition, CBS Publishers & Distributors Pvt. Ltd.	2007
2	Girdhari Lal, G. S. Siddappa, G. L. Tandon, "Preservation of Fruits and Vegetables",10th Reprint of 2nd Edition,Indian Council of Agricultural Research, New Delhi	2019
3	N.P.Singh, " Fruit and Vegetable Production", Oxford Book Company, India	2007
4	Bernard W. Minifie, "Chocolate, Cocoa and Confectionary: Science and Technology",3rd Edition, Springer Science and Business Media	1989
5	B.Srilaxmi, "Food Science", 8th Edition, New Age International Publishers	2018

Coursedetails

UNITS	S.NO	CONTENTS	LECTURE HOURS
I	1	Structural, Compositional and Nutritional aspects of fruits and vegetables. Physiological development: Growth, Maturation, Ripening and Senescence. Post harvest handling including controlled and modified storage.	4

	2	Techniques of processing and preservation of fruits and vegetables by refrigeration and freezing, canning and bottling, drying and dehydration.	**4**
II	**1**	Unit-II Technology of fruits and vegetable products: Juices and pulps, Concentrates and powders, Squashes and cordials. Beverage: Still and carbonated. James, Jellies and Marmalades. Preserves, candies and crystallized fruits.	**5**
		Technology of Tomato products: Puree, Paste, Ketchup, Sauce and Soup. Chutneys, pickles and other products. Characteristics and Industrial Production	**4**
III	**1**	Unit-III Spices: Composition, Structure and characteristics. Preservation and processing of major and minor spices of India; Whole spice, Spice powder, Paste and extracts, Spice oils and oleoresins.	**4**
	2	Composition, Structure , characteristics & processing of cashew nut and otherdry fruits. Important byproducts obtained and their utilization.	**3**
IV	**1**	Composition, Production and processing of Tea leaves: Black tea, Green tea and Oolong tea .Industrial Manufacturing of Tea CTC Tea, Instant tea. Iced Tea	**4**
	2	Production and processing of coffee cherries by wet and dry methods to obtain coffee beans, grinding, storage and preparation of brew, Soluble /Instant coffee, Use of chicory in coffee, Decaffeinated coffee.	**4**
V	**1**	Production, processing and chemical composition of cocoa beans. Cocoa Processes: Cleaning, roasting, alkalization, cracking and fanning, Nib grinding for cocoa liquor, cocoa butter and cocoa powder. Plastic range and Plasticity of fats	**4**
	2	Manufacturing process for chocolate: Ingredients, Mixing, Refining, Conching, Tempering, Moulding etc. to obtain chocolate slabs, chocolate bars. Significance of these processes. Enrobed chocolates and other confectionary products.	**4**

SUBJECT CODE: KFT-055	COURSE TITLE: NUTRACEUTICALS & FUNCTIONALFOODS
EXAM DURATION: 3.0 HOURS	SEMESTER : V
L: T: P :: 3 : 0 : 0 CREDITS:3	PRE REQUISITES: NIL

OBJECTIVES:

- **To impart the concept of nutraceuticals and functional ingredients in foods**
- **To determine their role in health and disease prevention.**
- **To study the usefulness and effects of Probiotics & Prebiotics in gastrointestinal health.**

COURSE OUTCOME:

After successful completion of the course the students will be able to:

1. Acquire knowledge on various bio molecules showing health benefits.
2. Understand various physiological and biochemical aspects of life threatening and chronic diseases.
3. Apply their knowledge regarding extraction, isolation, characterization and application of nutraceuticals infood industries.
4. Understand the manufacturing of functional foods and nutraceuticals.
5. Identify various aspects about safety, quality and toxicology of food products including nutraceutical and functional foods.
6. Understand the legal and regulatory aspects of Nutraceuticals and Functional foods

COs	Bloom's Level	POs 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) "-": No Correlation											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	K3	2	3	3	3	-	3	-	-	-	-	-	2
CO2	K2	2	2	3	2	-	3	-	-	-	-	-	2
CO3	K2	2	3	2	3	-	3	-	-	-	-	-	2
CO4	K3	2	-	3	2	-	3	-	-	-	-	-	2
CO5	K3	2	3	3	3	-	3	-	-	-	-	-	2
CO6	K3	2	2	3	2	-	3	-	-	-	-	-	2
Average		2.0	2.2	2.8	2.5	-	3.0	-	-	-	-	-	2.0

REFERENCEBOOKS

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1	Robert E.C., T C. Wallace, Handbook of Nutraceuticals & Functional Foods, 2nd Edition, CRC Press	2016
2	Robert E.C. Wildman, Denis M. Medeiros, Advanced Human Nutrition, CRC Press	2014
3	Shubhangini A. Joshi Nutrition and Dietetics,5th Edition, Tata McGraw-Hill	2021
4	N. Shakuntala Manay and M.Swamy: Food Facts & Principles, New Age International Publishers	2008

Course details

UNITS	S.NO	CONTENTS	LECTURE HOURS
I	1	Defining nutraceuticals and functional foods, Nature, type and scope.	3
	2	Nutraceuticals and functional foods applications and their health benefits, classification based on chemical and biochemical nature with suitable and relevant descriptions.	5

II	**1**	Nutraceuticals for specific situation such as cancer, heart dieases, stress, Osteoartehritis, hypertension etc	**4**
	2	Antioxidants and other phytochemicals, iso-flavones, lycopenes, their role in nutraceuticals and functional foods, dietary fibers and complex carbohydrates as functional food ingredients.	**4**
III	**1**	Protein as a functional food ingredients, Probiotic foods and their functional role, Herbs as functional foods, health promoting activity of common herbs.	**5**
	2	Cereals products as functional foods- Oats, Wheat bran, rice bran etc.	**4**
IV	**1**	Functional vegetable products, oil seeds and sea foods. Coffee, tea and other beverages as functional foods/ drinks and their protective effects.	**5**
	2	Effects of processing and storage and interaction of various environmental factors on the potentials of such foods.	**4**
V	**1**	Marketing and Regulatory issues for functional foods and nutraceuticals.	**4**
	2	Recent developments and advances in the area of nutraceuticals and functional foods.	**4**

SUBJECT CODE: KFT-056	COURSE TITLE: TECHNOLOGY OF FOOD BEVERAGES
EXAMINATION DURATION: 3Hrs.	SEMESTER: V
L:T: P : : 3: 0: 0 CREDITS: 3	PRE-REQUISITE: NIL

OBJECTIVES:

- **To provide basic knowledge of Food beverages and their classification**
- **To provide knowledge of operations & machinery of beverage industry.**
- **To provide knowledge regarding Quality control & Safety of various beverages**

COURSE OUTCOME:

After successful completion of the course the students will be able to:

1. Characterize the different categories of food beverages.
2. Understand the technology and machinery of carbonated beverage manufacturing.
3. Understand the technology and machinery of non carbonated fruit based beverage manufacturing.
4. Understand the Quality aspects /relevant parameters of beverages and their estimation.
5. Understand the packaging aspects of different types of food beverages.
6. Understand the Waste management in beverage industries.

COs	Bloom's Level	POs 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) "-": No Correlation											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	K3	1	1	2	1	-	3	-	-	-	-	-	-
CO2	K2	2	1	3	1	-	-	-	-	-	-	-	-
CO3	K2	1	2	3	2	-	3	-	-	-	-	-	-
CO4	K3	2	2	3	2	-	3	-	-	-	-	-	2
CO5	K3	-	2	3	2	-	3	-	-	-	-	-	2
CO6	K3	3	2	3	2	-	3	-	-	-	-	-	2
Average		1.5	1.7	2.8	1.7	-	2.5	-	-	-	-	-	1.0

REFERENCEBOOKS:

S.No	Name of Authors / Books / Publishers	Year of Publication/ Reprint
1.	L.F.Green: Developments in Soft Drinks Technology-I, Applied Science PublishersLtd., London	1978
2.	Norman N.Potter, Joseph H.Hotchkiss, "Food Science", 5th Edition, CBS Publishers& Distributors Pvt. Ltd.	2007
3.	A.G.H. Lea, J.R.Piggot :Fermented Beverage Production, 2nd Edition, Springer-Science, Business Media, B.V	2003
4.	De Sukumar: Outlines of Dairy Technology, 46th Edition, Oxford University Press	2019

COURSEDETAILS:

UNITS	S.NO.	CONTENTS	LECTURE HOURS
I	1	Introduction, Definition and classification of Beverages, Beverage industry in India, Traditional beverages	4
	2	Ingredients and their role.Effect of different ingredient on the quality of beverages; Preparation ofsyrups.	4

II	**1**	Water quality, treatment and fortification process, Bottling, Packaging, storage and transportation. Specifications of Packaged water as per FSSAI.	**4**
	2	Manufacturing technology of mineral water and Flavoured carbonated Beverages. Quality Parameters of the prepared products.	**4**
III	**1**	Fruit beverages; Definition and preparation of squash, cordial, nectar, crush.	**4**
	2	Alcoholic beverages, Milk beverages, Preparation of different beverages	**5**
IV	**1**	Selection and economics of different beverages packaging materials	**3**
	2	Selection, operation and maintenance of beverage machines / equipments, Automation in beverage industries.	**4**
V	**1**	Quality control and Safety (HACCP Plan) in beverage industries, Waste management in beverage industries	**4**
	2	Sensory evaluation of beverages, Chemical and microbiological analysis of different beverages.	**4**

SUBJECTCODE: **BCH554**	COURSETITLE: **CHEMICAL REACTION ENGINEERING LAB**
EXAMINATION DURATION: 2Hrs	SEMESTER: **V**
L: T: P::0 : 0 :2 CREDITS: 1	PRE-REQUISITE: **MEB and Chemical Thermodynamics**

OBJECTIVES:

- To impart knowledge about the basic fundamental principles of chemical reaction engineering by performing different experiments.
- To make them correlate theory and practical process by experimentation.
- To provide them practical know how about different types of reactors and also the concept of non-ideal flow.

COURSE OUTCOME:

1. Determine the order and rate constant of chemical reactions in various reactor setups (PFR, CSTR, Batch).
2. Analyze the esterification reaction and determine the order and rate constant at room temperature.
3. Understand and operate adiabatic batch reactors for reaction studies.
4. Evaluate reaction equilibrium constants for the reaction of acetic acid with ethanol.
5. Study residence time distribution and plot F-Curve and C-Curve in stirred tank and plug flow reactors.
6. Calculate the energy of activation for a given chemical reaction.

CO/PO	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-
CO-1	3	3	2	2	2	1	1	1	1	1	1	2
CO-2	3	3	2	2	2	1	1	1	1	1	1	2
CO-3	2	2	3	2	2	1	1	1	1	1	1	2
CO-4	3	3	2	2	2	1	1	1	1	1	1	2
CO-5	3	3	2	2	2	1	1	1	1	1	1	2
CO-6	3	3	2	2	2	1	1	1	1	1	1	2

REFERENCEBOOKS

S.No	NAME OF AUTHORS /BOOKS/ PUBLISHERS	YEAR OFPUBLICATION/ REPRINT
1	Smith,J,M,"ChemicalEngineeringKinetics",3rdEdition,McGraw-Hill (1990).	1990
2	Levenspiel,O.,"ChemicalReactionEngineering",3rdEdition,JohnWiley, (1998).	1998
3	FoglerH.S.,Elementsof ChemicalReactionEngineering,4th edition,Prentice Hallof India,(2008)	1997

LISTOFEXPERIMENTS:

1. To determine the order and rate constant of saponification reaction at room temperature in a PFR.
2. To determine the order and rate constant of saponification reaction at room temperature in a CSTR
3. To determine the order and rate constant of saponification reaction at room temperature in a Batch reactor.
4. To determine the order and rate constant of esterification reaction at room temperature.
5. Study and operation of an adiabatic batch reactor
6. Study of a reversible reaction in a batch reactor
7. To determine reaction equilibrium constant of reaction of acetic acid with ethanol.
8. To study the residence time distribution in a stirred tank reactor and plot the F-Curve and C-Curve for given Reactor.
9. To study the residence time distribution in a plug flow reactor and plot the F-Curve and C-Curve for given Reactor.
10. To determine the energy of activation of a given chemical reaction**.**

SUBJECT CODE: KFT-551	COURSE TITLE: FOOD PRESERVATION LAB
EXAM DURATION: 2.0 HOURS	SEMESTER : V
L: T: P ::0 : 0 : 2 CREDITS: 1	PRE REQUISITES: NIL

OBJECTIVES:

- **To provide basic insight into the practical aspects of the principles of food preservation**
- **To provide knowledge to ascertain the efficacy of these processes**

COURSE OUTCOME:

After successful completion of the course the students will be able to:

1. Understand the composition and characteristics of fresh produce
2. Understand the causes of post harvest losses and basic principles of preservation.
3. Understand the causative factors of spoilage and means to prevent this spoilage
4. Understand the application of various processes towards preservation of foods.
5. Understand to determine the efficacy of preservation methods.
6. Understand the usage and limits of additives in foods as preservatives.

COs	Bloom's Level	POs 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) "-": No Correlation											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	K3	2	1	1	-	2	2	-	-	-	-	-	2
CO2	K2	3	2	3	2	2	3	-	-	-	-	-	2
CO3	K2	1	2	2	2	2	3	-	-	-	-	-	1
CO4	K3	2	2	2	3	3	3	-	-	-	-	-	2
CO5	K4	2	2	2	2	3	-	-	-	-	-	-	1
CO6	K3	2	2	2	2	3	2	-	-	-	-	-	2
Average		2.0	1.8	2.0	1.8	2.5	2.2	-	-	-	-	-	1.71.7

REFERENCEBOOKS

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/REPRINT
1	Norman N.Potter, Joseph H.Hotchkiss, " Food Science", 5th Edition, CBS Publishers & Distributors Pvt. Ltd.	2007
2	Girdhari Lal, G. S. Siddappa, G. L. Tandon, "Preservation of Fruits and Vegetables",10th Reprint of 2nd Edition,Indian Council of Agricultural Research, New Delhi	2019
3	M.Shafiur Rahman, "Handbook of Food Preservation" 2nd Edition, CRC Press	2007
4	S.Ranganna, "Handbook of Analysis and Quality Control for Fruit and Vegetable Products". S, Tata McGraw-Hill Education	1986
5	B.Srilaxmi, "Food Science", 8th Edition, New Age International Publishers	2018

Course details

S.No	Name of Experiment
1	Study of Post harvest Handling and Treatment of Fruits and Vegetables
2	Study of Post Harvest Handling of Grains, Pulses and Milk
3	Study of Preservation of fresh produce by Freezing
4	Study of Preservation of Foods by High Temperature [Canning/ Bottling]
5	To check the efficacy/ adequacy of Blanching treatment
6	Study of preservation of foods by Drying / Dehydration
7	Study of Osmotic Concentration / Osmo Air Drying as a means of Food Preservation
8	Study of Fermentation as a means of Food Preservation
9	Study of preservation of foods by use of Chemical Preservatives
10	Study of Microwave Pasteurization and Sterilization of Foods

SUBJECT CODE: KFT-552	COURSE TITLE: FOOD PRODUCTS LAB-I
EXAM DURATION: 2.0 HOURS	SEMESTER : V
L: T: P ::0 : 0 : 2 CREDITS: 1	PRE REQUISITES: NIL

OBJECTIVES:

- **To provide basic insight into the practical aspects of preservation of fresh produce**
- **To provide knowledge about the preparation of Value Added Food Products**
- **To provide insights into testing of Quality parameters involved.**

COURSE OUTCOME:

After successful completion of the course the students will be able to:

- Understand the technologies for Value Addition of fresh produce
- Understand the preparation of Value added items from fruits and Vegetables.
- Understand the preparation of Value added items from Cereals and spices.
- Understand the defects encountered in such products and preventive measures to be adopted.
- Understand the causative factors of spoilage and means to prevent this spoilage
- Understand the product quality with reference to standard specifications.

COs	Bloom's Level	POs 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) "-": No Correlation											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	K4	2	1	1	-	2	2	-	-	-	-	-	2
CO2	K3	3	2	3	2	2	3	-	-	-	-	-	2
CO3	K3	1	2	2	2	2	3	-	-	-	-	-	1
CO4	K4	2	2	2	3	1	3	-	-	-	-	-	2
CO5	K4	1	3	3	3	3	3	-	-	-	-	-	2
CO6	K4	1	1	2	2	-	1	-	-	-	-	-	2
Average		1.7	1.8	2.2	2.0	1.7	2.5	-	-	-	-	-	1.8

REFERENCEBOOKS

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1	Norman N.Potter, Joseph H.Hotchkiss, " Food Science", 5th Edition, CBS Publishers & Distributors Pvt. Ltd.	2007
2	Girdhari Lal, G. S. Siddappa, G. L. Tandon, "Preservation of Fruits and Vegetables",10th Reprint of 2nd Edition,Indian Council of Agricultural Research, New Delhi	2019
3	Vijay Sethi, Shruti Sethai , B.C. Dekka , "Processing of Fruits and Vegetables for Value Addition", Indus Publishing Company	2006
4	S.Ranganna, "Handbook of Analysis and Quality Control for Fruit and Vegetable Products". S, Tata McGraw-Hill Education	1986
5	B.Srilaxmi, "Food Science", 8th Edition, New Age International Publishers	2018

Course details

S.No	Name of Experiment
1	Studies on Preparation of Jam, Jelly and Marmalade and their quality analysis
2	Studies on preparation of Fruit Leather and its quality analysis
3	Studies on preparation of Candies and Preserves and their quality analysis
4	Studies on preparation of Tomato Products: Tomato Sauce, Tomato Ketchup etc. and their quality analysis
5	Studies on preparation of fruit squash and cordial and their quality analysis
6	Studies on preparation of Pickles with salt and sugar
7	Studies on preparation of Fermented Pickles
8	Studies on preparation of Rice Products
9	Studies on preparation of Corn Products
10	Studies on preparation of Spice pastes

B.TECH III YEAR VI SEMESTER FOOD TECHNOLOGY COURSE

SUBJECT CODE:BCH 604	**COURSE TITLE:INSTRUMENTATION & PROCESS CONTROL**
EXAMDURATION:3 Hrs	**SEMESTER:VI**
L: T: P:: 3 : 1: 0 CREDITS:4	**PRE-REQUISITE:NIL**

OBJECTIVES:

- To enable the students to understand the fundamentals of process control
- To provide knowledge about various types of measuring instruments used in different types of processes
- To impart knowledge about characteristics of different types of controllers for controlling a process and process automation

COURSE OUTCOME:

1. Understand the elements, classifications, and characteristics of measuring instruments.
2. Apply principles of industrial instruments for temperature, flow, level, and composition measurement.
3. Analyze and model the dynamic behavior of first and second-order systems in process control.
4. Design and evaluate process control systems, including open and closed-loop control.
5. Implement advanced control strategies such as cascade, feed forward, and ratio control.
6. Understand and apply the principles of electro-pneumatic transducers and pneumatic control valves.

CO/PO	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	3	2	1	1	2	1	1	1	1	1	1	2
CO-2	3	3	2	2	3	1	1	1	1	1	1	2
CO-3	3	3	2	3	3	1	1	1	2	2	1	3
CO-4	3	3	3	3	3	2	1	1	2	2	1	3
CO-5	3	3	3	3	3	2	1	1	2	2	1	3
CO-6	3	2	2	2	3	1	1	1	1	1	1	2

REFERENCEBOOKS

S.NO	**NAMEOFAUTHORS/BOOKS/PUBLISHERS**	**YEAR OF PUBLICATION / REPRINT**
1	Eckman, D.P., Industrial Instrumentation, Wiley Eastern Ltd., NewYork.	1990
2	Jain, R.K., Mechanical and Industrial Measurements, Khanna Publishers.	1995
3	Coughnaowr, D.R. ,"Process Systems Analysis and Control", McGraw-Hill, Inc.	1991
4	Stephanopolous,G.,"Chemical Process Control", Prentice-Hall.	1984

COURSEDETAILS

UNITS	**CONTENTS**	**LECTURE HOURS**
I	Elements of measurement, functions and general classifications of measuring instruments.Indicatingandrecordingtypeofinstruments.Elementsofmeasuringinstruments,static And dynamic characteristics of measuring instruments.	8

II	Principle of operation,constructionandapplicationofimportantindustrialinstrumentsfor The measurement of temperature, flow, liquid level and composition. Principles and construction of electro-pneumatic transducer and pneumatic control valve.	**9**
III	Instruments for Measurement of Flow rate, level &Viscosity, Variable Area &variable headflowmeters.IntroductiontoProcesscontrolsystems,Regulator&Servocontrol,Negative &Positive Feedback Control	**10**
IV	Dynamic Modeling of a Process, Dynamic behavior of First order systems and First order systems in series. Dynamic behavior of second & higher order system for various kind of inputs, Linearization of non linear system, Transportation &Transfer Lag.	**9**
V	Block and physical diagrams of control system. Open and closed loop control systems. Characteristics of measuring elements, controllers and final control elements. Mode of Control actions. Advanced control strategies, cascade control, Feed Forward &Feed backward control, Ratio control.	**9**

SUBJECT CODE: KFT-601	COURSE TITLE: BAKERY TECHNOLOGY
EXAM DURATION: 3.0 HOURS	SEMESTER : VI
L: T: P ::3 : 1 : 0 CREDITS:4	PRE REQUISITES: NIL

OBJECTIVES:

- To impart fundamental knowledge of bakery products
- To impart basic knowledge about machinery and equipments involved
- To give brief introduction of recent developments in bakery product technology.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

1. To acquire knowledge of bakery products.
2. Understand the importance and role of ingredients in bakery industry
3. Understand the technology involved in production of Bread, Biscuits, Cakes and other baked products.
4. Understand about the machinery involved in manufacture of these products.
5. Understand the types of defects in bakery products and their remedies
6. Understand the packaging requirements of bakery products

COs	Bloom's Level	POs 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) "-": No Correlation											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	K3	2	2	2	1	2	-	-	-	-	-	-	-
CO2	K2	3	1	2	1	3	1	-	-	-	-	-	1
CO3	K2	3	2	1	2	3	1	-	-	-	-	-	1
CO4	K3	2	2	3	2	1	1	-	-	-	-	-	2
CO5	K3	2	2	2	2	1	1	-	-	-	-	-	2
CO6	K3	2	2	2	2	1	1	-	-	-	-	-	2
Average		2.3	1.8	2.0	1.7	1.8	0.9	-	-	-	-	-	1.3

REFERENCEBOOKS

S.NO	NAMEOFAUTHORS/BOOKS/PUBLISHERS	YEAR OF PUBLICATION /REPRINT
1	Pyler & Gorton: Baking Science and Technology, 4th Edition, Sosland Pub Co.	2009
2	Samuel A. Matz: Bakery Technology and Engineering (Paperback) Springer US	2019
3	Samuel A. Matz: Cookie and Cracker Technology, 3rd Edition, Van Nostrand Reinhold Inc.,U.S	1993
4	H. Faridi: The Science of Cookie and Cracker Production, Chapman & Hall	1997

Course details

UNITS	S.NO	CONTENTS	LECTURE HOURS
I	1	Wheat flour and wheat flour treatments, Grade of flour, constituents of flour, ageing of flour, Tests for flour quality.	4
	2	Yeast: Characteristics, Preparation, Handling & Storage, Adequacy for use in bakery industry.	2
	3	Ingredients, Technology and quality parameters for baked products: Bread, Biscuits and cakes.	3
II	1	Bakery equipment and machinery. Different types of Mixers, kneaders and cutters. Different types of ovens.	4

	2	Packaging machinery for bread and biscuits. Quality control in bakery industry. Quality control of raw materials. Quality control of finished products. Quality control of packaging materials.	**4**
III	**1**	Technology of bread making - Different methods, process steps and their significance.	**5**
	2	Characteristics of good bread. Defects in bread their causes and remedies.	**4**
IV	**1**	Technology of Cakes Manufacture. Different cake making processes. Sugar batter method, Flour batter method, Modified sugar batter method Whipping method, Blending method etc. Process steps and their significance.	**5**
	2	Importance of baking time and temperature. Recipe balancing. Defects in cakes, their causes and remedies.	**3**
V	**1**	Biscuits - Definition and types. Short dough, hard Dough and Fermented dough biscuits.	**3**
	2	Cookies - Types of cookies and their manufacture.	**2**
	3	Cream biscuits - Process steps and their significance. Defects in biscuits their causes and remedies.	**3**

SUBJECT CODE: KFT-602	COURSE TITLE: DAIRY TECHNOLOGY
EXAM DURATION: 3.0 HOURS	SEMESTER : VI
L: T: P ::3 : 1 : 0 CREDITS: 4	PRE REQUISITES: NIL

OBJECTIVES:

- To impart fundamental knowledge of the technologies of processing of milk and milk products
- To impart basic knowledge of the machinery and equipments involved.
- To explain the hygiene and sanitation practices in milk plant.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

1. Understand the composition and properties of fluid milks.
2. Understand the processes related to storage, processing and distribution of milk.
3. Understand the functioning of various equipments related to dairy industry
4. Understand the technology and principles involved in manufacture of milk products.
5. Understand the Byproducts of Dairy Industry and their effective utilization
6. Have knowledge regarding hygiene and sanitation practices in the milk and milk products industry.

COs	Bloom's Level	POs 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) "-": No Correlation											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	K3	3	1	-	-	1	1	-	-	-	-	-	1
CO2	K3	3	2	2	3	2	3	-	-	-	-	-	2
CO3	K2	3	2	-	-	2	3	-	-	-	-	-	2
CO4	K3	3	2	2	2	2	3	-	-	-	-	-	2
CO5	K3	3	2	2	2	2	2	-	-	-	-	-	2
CO6	K2	3	1	2	1	2	1	-	-	-	-	-	1
Average		3.0	1.7	1.3	1.3	1.8	2.2	-	-	-	-	-	1.7

REFERENCEBOOKS

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1	De Sukumar: Outlines of Dairy Technology, 46th Edition, Oxford University Press	2019
2	Aneja, Mathur, Chandan & A. K. Bannerji: Technology of Indian Milk Products: Dairy India Publication	2002
3	Shivashraya Singh: Dairy Technology: Vol.01: Milk and Milk Processing,New India Publishing Agency	2013
4	Meghwal. Goyal and Chavan: Dairy Engineering: Advanced Technologies andtheir applications, Apple Academic Press, Canada.	2017

COURSEDETAILS

UNITS	S.NO	CONTENTS	LECTURE HOURS
I	1	Fluid Milk: Composition of milk and factor affecting it. Physico-chemical characteristics of milk and milk constituents.	5
	2	Production and collection, cooling and transportation of milk. Packaging storage and distribution of pasteurized milk.	4
	1	Whole, Standardized, Toned, Double toned and skim milk. Test for milk quality and Adulteration.	3

II	**2**	UHT processed milk, flavored, Sterilized milk. Cleaning and sanitization of dairy equipments.	**3**
	3	Definition, Classification, Composition and physico-chemical properties of cream. Production processes and quality control.	**4**
III	**1**	Butter: Definition, Classification, Composition and methods of manufacture, Packaging and storage. Butter oil/Ghee.	**4**
	2	Dairy based Ice cream: Definition, Classification and Composition, Constituents and their role. Preparation of mixes and freezing of Ice cream, Overrun, Judging, Grading, and defects of Ice cream.	**4**
IV	**1**	Evaporated and Condensed milk: Method of manufacture, Packaging and storage. Defects, Causes, and prevention.	**3**
	2	Roller and Spray Drying of milk solids. Instantization. Flow ability, Dustiness, Reconstituability, Dispersability, Wetability, Sink ability and appearance of milk powders.	**5**
V	**1**	Byproducts of Dairy Industry and their effective utilization.	**3**
	2	Manufacture of casein, Whey protein, Lactose from milk and their use in formulated foods. Quality Control tests in Dairy industry.	**4**

SUBJECT CODE: KFT-061	COURSE TITLE: TECHNOLOGY OF ANIMAL FOODS
EXAMINATION DURATION: 3Hrs.	SEMESTER: VI
L: T: P : : 3: 0: 0 CREDITS: 3	PRE-REQUISITE: NIL

OBJECTIVES:

- To give the students an understanding of the nutritional value, processing, and storage characteristics of meat, poultry and fish as food
- To provide knowledge regarding the manufacture of Food Products from these sources
- To provide information about quality control and by-product utilization of the meat, poultry and fish industry.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

1. Understand the ante-mortem and postmortem factors affecting the quality of meat produced.
2. Understand the processing and preservation technology involved in industrial manufacture of meat.
3. Understand the processing technology involved in industrial manufacture of poultry and fish products
4. Understand the composition , properties of egg and technology of egg based products
5. Be able to understand various quality and safety aspects of meat, poultry and fish processing industry.
6. Understand the technology of by-product utilization in meat, poultry and fish industry

COs	Bloom's Level	POs 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) "-": No Correlation											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	K2	2	1	1	1	-	3	-	-	-	-	-	-
CO2	K2	2	1	2	1	-	3	-	-	-	-	-	-
CO3	K3	1	1	1	2	-	3	-	-	-	-	-	-
CO4	K2	2	2	2	1	-	3	-	-	-	-	-	-
CO5	K3	2	2	2	1	-	3	-	-	-	-	-	-
CO6	K3	2	2	2	1	-	3	-	-	-	-	-	-
Average		1.8	1.5	1.7	1.2	-	3.0	-	-	-	-	-	-

REFERENCEBOOKS:

S.No	Name of Authors / Books / Publishers	Year of Publication /Reprint
1.	G.J.Mountney, Poultry Products Technology , 3rd Edition, CRC Press	2017
2.	R. A. Lawrie, David Ledward: Meat Science, Wood head Publishing	2017
3.	F.Toldra, Lawries's Meat Science, 9th Edition, . Woodhead Publishing	2022
4.	Bremner HA , Safety and Quality Issues in Fish Processing, Woodhead Publishing	2002

COURSEDETAILS:

UNITS	S.NO.	CONTENTS	LECTURE HOURS
I	1	Ante-mortem examination of meat animals, Scientific slaughtering; Meat cuts and portions of meat,Post mortem changes in meat	4
	2	Conversion of muscle to meat; Colour of meat; Composition and nutritional value, Meat microbiology and safety.	4
II	1	Meat processing curing and smoking; Fermented meat products (sausages and sauces);	4
	2	Frozen meat & meat storage. Beef Mutton, Pork Sausages and other meat products.	4
III	1	Poultry processing, Canning of poultry products.	4
	2	Structure, composition and Nutritional and Functional characteristics of Egg. Causes of deterioration of quality of egg, Preservation and Processing of Egg.	4
	3	Manufacturing of egg white, Egg yolk and Whole Egg solids/powder.	2

IV	**1**	Classification of fresh water fish and marine fish; Commercial handling, storage and transport of raw fish. Average composition of fish; Freshness criteria, quality assessment of fish; Spoilage of fish.	**3**
	2	Methods of processing and preservation of fish- Canning, Freezing, Drying, Smoking and Curing. Fish products – fish meal, fish protein concentrate, fish liver oil, fish sauce and surimi; Fish processing industries in India.	**5**
V	**1**	Meat plant hygiene – GMP and HACCP. By-products from meat industries and their utilization; Meat industries in India. Production of Chitin, Chitosan; Production of non-food items from fish processing wastes.	**4**
	2	Byproduct Utilization – commercial processing of lecithin and other egg solids, Utilization of egg-derived products as food ingredients; Fertilizer from shells.	**4**

SUBJECT CODE: KFT-062	COURSE TITLE: TRADITIONAL & FERMENTED FOODS
EXAMINATION DURATION: 3Hrs.	SEMESTER: VI
L: T: P: : 3: 0: 0 CREDITS: 3	PRE-REQUISITE: NIL

OBJECTIVES:

- To impart student basic knowledge of the traditional Indian sweets, savory and snack products
- To impart knowledge regarding their manufacturing technology and Quality Aspects
- To give them understanding of fermented food products, technology of production and quality aspects.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

1. Understand the concept of traditional Indian food products.
2. Understand the processing technology and quality of traditional Indian food products.
3. Understand the importance and techniques of fermentation in food industry.
4. Understand the technology of different type food products produced by fermentation
5. Understand the direct or indirect use of microorganism in the food production industry.
6. Understand the quality &safety requirements involved in manufacture of fermented food products.

COs	Bloom's Level	POs 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) "-": No Correlation											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	K3	2	1	2	1	-	3	-	-	-	-	-	-
CO2	K2	3	1	3	1	-	-	-	-	-	-	-	-
CO3	K2	3	2	3	2	-	3	-	-	-	-	-	-
CO4	K3	3	2	3	2	-	3	-	-	-	-	-	2
CO5	K3	3	2	2	2	-	-	-	-	-	-	-	2
CO6	K3	3	2	2	2	-	3	-	-	-	-	-	2
Average		2.8	1.7	2.5	1.7	-	2.0	-	-	-	-	-	1.0

REFERENCEBOOKS:

S.No	Name of Authors / Books / Publishers	Year of Publication/ Reprint
1.	K.H. Steinkrus : Handbook of Indigenous Fermented Foods, CRC Press,	2018
2.	De Sukumar: Outlines of Dairy Technology, 46th Edition, Oxford University Press	2019
3.	Prescott & Dunn Industrial Microbiology,4th Ed. CBS Publishers and Distributors	2005
4.	L.E. Casida Industrial Microbiology, 2nd Edition, New Age International Publishers	2019

COURSEDETAILS:

UNITS	S.NO.	CONTENTS	LECTURE HOURS
I	1	Indian traditional sweet, savory and snack food products: Sweetmeats, Namkins, Papads , Idli and Dosa.	4
	2	Raw materials, Role of Ingredients and preparation.	4
II	1	Preparation and Maintenance of Microbial cultures for food fermentations. Industrially important cultures, Inoculum Development for industrial fermentations	4
	2	Lactic acid bacteria-activities and health-promoting effects. Mushrooms: Cultivation and preservation.	4
III	1	Fermented Dairy Products: Cheeses, Curd and Yoghurt, Butter milk and the fermented milks. Spoilages and defects of fermented dairy products and their control.	5

	2	Preparation and processing of Fermented meat and fish products.	**3**
IV	**1**	Fermentative Production of Alcoholic Beverages as Beer, Wines, Cider and Vinegar. Raw materials,Role of Ingredients, Process, Quality Factors, Common faults and Remedies.	**5**
	2	Principles and Process involved in the making of Fermented Vegetables (Pickles).	**3**
V	**1**	Production of Baker's Yeast, Microbial Proteins and fats, Food enzymes, and Food additives.	**4**
	2	Oriental fermented foods. Soya Sauce, Koji, Tempeh, Miso, Natto, Tofu, Minchin etc.	**4**

SUBJECT CODE: KFT-063	COURSE TITLE: SPECIALITY FATS FOR FOOD INDUSTRY
EXAM DURATION: 3.0 HOURS	SEMESTER : VI
L: T: P :: 3 : 0 : 0 CREDITS: 03	PRE REQUISITES: NIL

OBJECTIVES:

- To provide knowledge about Bakery & Shortening Fats, Confectionery Fats
- To provide knowledge about Margarines, Spread Fats, and Fractionated Fats for better understanding of end products of Industry
- To disseminate knowledge on Functional Fats & their qualitative and sensory aspects
- To prepare passing out graduates for better job opportunities multi products food industry

COURSE OUTCOME:

After successful completion of the course the students will be able to:

1. Understand the basic fundamentals of specialty fats
2. Understand functions and applications of modified fats
3. Understand the sensory aspects of speciality fats
4. Apply knowledge to formulate different specialty fats
5. Analyze the process used for production of specialty fats
6. Analyze the specialty products from quality aspects

COs	Bloom's Level	POs 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) "-": No Correlation											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	K3	2	1	1	2	1	2	-	-	-	-	-	-
CO2	K2	2	1	1	2	1	2	-	-	-	-	-	-
CO3	K2	2	-	-	2	-	2	-	-	-	-	-	-
CO4	K3	3	2	2	3	3	2	-	-	-	-	-	-
CO5	K3	3	2	2	3	3	2	-	-	-	-	-	-
CO6	K3	3	2	2	3	3	2	-	-	-	-	-	-
Average		2.5	1.3	1.3	2.5	1.8	2.0	-	-	-	-	-	-

REFERENCEBOOKS

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1	Bailey's Industrial Oil and Fat, Edition 6 ;Vol-2,3 & 5 Edited by Feireidoon Shahidi	2005
2	H.B.W. Patterson , Hydrogenation of Oil & Fat , London Applied Science	1983
3	Manoj K. Gupta, "Practical Guide to Vegetable Oil Processing" AOCS publication	2007
4	Richard D.O. Brien , "Fats and oils formulating and processing for applications"., 2nd Edition CRC Press Florida Publication	2003

Course details

UNITS	S.NO	CONTENTS	LECTURE HOURS
I	1	**Introduction to Speciality Fats:** Definition of speciality fats, Plasticity of Fats and Plastic range, Functions and Applications of Shortening and Bakery Fats, Cocoa Butter: Physicochemical properties; Fatty acid composition; Cocoa Butter Alternatives, viz Cocoa Butter Equivalent (CBE), Cocoa Butter Substitutes (CBS), Cocoa Butter Replacer (CBR); Low Calorie Fats	8
		Functions and Applications of Modified Fats: Hydrogenation Reaction, Hydrogenated Fats, Inter-esterification; Random Inter-	8

II	**1**	esterification, Directed Inter-esterification,, Margarine, Fat Crystallization into various polymorphic forms, Palm fractions, Palm Kernel Oils; application of Palm oil in solid margarin, Stearin, Palm Kernel Stearin in confectionery industry, Confectionary lipids, Fats and Oils for Bakery Products, Margarines & Spreads	
III	**1**	**Flavours and Sensory Evaluation of Fats:** Chemistry, function and applications of various additives such as preservatives, antioxidants, emulsifiers, humectants, stabilizers used in formulations, Flavour Components of fats, Flavours & Sensory aspects of fats	**8**
IV	**1**	**Modification of Fats:** Hydrogenation Process, Trans Fat alternatives and challenges, Margarine Processing Plant, Palm Oil Fraction , Crystallization of Oils & Fats	**8**
V	**1**	**Analytical Aspects:** Dilatometry- theory and practice, Solid Fat Content, Solid Fat Index, determination of Peroxide Value Anisidene value, Totox value, Determination of Polar Components, Quality control and assurance using GC,GC-MS-MS, LC-MS-MS, Rancimat and detection of formation of MCPD	**8**

SUBJECTCODE:BCH 655	**COURSE TITLE: INSTRUMENTATION & PROCESS CONTROL LAB**
EXAMINATION DURATION:2Hrs	**SEMESTER:VI**
L: T: P:: 0 : 0 :2 CREDIT: 1	**PRE-REQUISITE:NIL**

OBJECTIVES:

- To make students aware of working of different process control instruments through hands-on training.
- To make students to correlate theory and practical process control through principles, fundamental concepts and by experimentation.

COURSEOUTCOME:

1. Calibrate and validate thermocouples, pressure gauges, and various flow meters.
2. Estimate viscosity using Redwood, Saybolt, and Ostwald viscometers.
3. Calibrate and operate pH meters for accurate pH measurement.
4. Analyze the response and time constant of thermocouples in different conditions.
5. Study and interpret the dynamic response of single and multi-tank systems.
6. Evaluate the performance of PI/PID pneumatic and electronic controllers.

CO/PO	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	3	2	2	2	2	1	1	1	1	1	1	2
CO-2	2	2	2	1	2	1	1	1	1	1	1	2
CO-3	3	2	2	2	2	1	1	1	1	1	1	2
CO-4	2	2	2	1	2	2	1	1	1	1	1	2
CO-5	2	2	2	1	2	2	1	1	1	1	1	2
CO-6	2	2	2	1	2	2	1	1	1	1	1	2

REFERENCEBOOKS:

S.NO	NAMEOFAUTHORS/BOOKS/PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1	Eckman,D.P., Industrial Instrumentation, Wiley Eastern Ltd.,NewYork.	1990
2	Jain, R.K., Mechanical and Industrial Measurements, Khanna Publishers.	1995
3	Coughnaowr,D.R.,"ProcessSystemsAnalysisandControl",McGraw-Hill,Inc.	1991
4	Stephanopolous, G.," Chemical Process Control", Prentice-Hall.	1984

COURSEDETAILS:

LISTOFEXPERIMENTS:

1. Calibration of thermocouple/Bimetallic thermocouple/Resistance thermocouple.
2. Calibration of Pressuregauge/Pneumaticpressurerecorder/Differentialpressurerecorder.
3. Calibration of Orificemeter/ Venturimeter/ Rotameter/ Gasflowmeter.
4. Estimation of viscosity by Redwood/Saybolt/Ostwald viscometer.
5. Calibration of Ph meter.
6. To study the response, time constant of thermocouple.
7. To study the response of a liquid level tank system
8. To study the response of a two tank non-interacting system
9. To study the response of a two tank interacting system.
10. To study the characteristics of a PI/PID pneumatic/electronic controller.

SUBJECT CODE: KFT-651	COURSE TITLE: BAKERY TECHNOLOGY LAB
EXAM DURATION: 2.0 HOURS	SEMESTER : VI
L: T: P ::0 : 0 : 2 CREDITS: 1	PRE REQUISITES: NIL

OBJECTIVES:

- To provide students with knowledge and skills necessary to produce quality baked goods**.**
- To provide knowledge regarding testing of quality parameters of baked products

COURSE OUTCOME:

After successful completion of the course the students will be able to:

1. Understand the application of baking ingredients
2. Understand the role of equipment and tools in baking technology.
3. Understand the formulation calculations involving scaling and measuring ingredients.
4. Understand the manufacturing process of different bakery products.
5. Understand the defects encountered in bakery products and preventive measures to be adopted.
6. Analyze the quality parameters of prepared bakery products.

COs	Bloom's Level	POs 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) "-": No Correlation											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	K4	2	2	2	1	2	-	-	-	-	-	-	-
CO2	K3	3	1	2	1	3	1	-	-	-	-	-	1
CO3	K4	3	3	1	2	3	1	-	-	-	-	-	1
CO4	K4	3	3	1	2	3	1	-	-	-	-	-	1
CO5	K4	3	3	2	2	3	1	-	-	-	-	-	1
CO6	K4	3	3	2	1	3	1	-	-	-	-	-	1
Average		2.8	2.5	1.7	1.5	2.8	0.8	-	-	-	-	-	0.8

REFERENCE BOOKS

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1	S.C Dubey: Basic Baking: Science and Craft, Publisher	1980
2	Betty Crocker: Baking Basics, Wiley	2009

Course details

S. No.	Name of Experiment
1	Preparation of Bread/ Test Baking.
2	Preparation of Sweet Buns
3	Preparation of Pizza base.
4	Preparation of Biscuits
5	Preparation of Nan-Khatai
6	Preparation of Cookies
7	Preparation of Cakes (Eggs/ Eggless)
8	Preparation of Pastries
9	Preparation of Laminated and Puffed products

SUBJECT CODE: KFT-652	COURSE TITLE: DAIRY TECHNOLOGY LAB
EXAM DURATION: 2.0 HOURS	SEMESTER : VI
L: T: P ::0 : 0 : 2 CREDITS: 1	PRE REQUISITES: NIL

OBJECTIVES:

- To provide knowledge about the quality of milk and milk products.
- To depict the detection of adulterants in milk and milk products.
- To provide knowledge regarding the quality of dairy products.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

1. Understand basic principles of milk quality analysis.
2. Understand different methods of raw milk quality analysis.
3. Understand the significance and role of various equipments used in dairy industry
4. Understand detection methods of adulterants in milk.
5. Understand the preparation of various dairy products
6. Analyze the quality parameters of different dairy products.

COs	Bloom's Level	POs 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) "-": No Correlation											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	K4	3	1	-	-	1	1	-	-	-	-	-	1
CO2	K3	3	2	2	3	2	3	-	-	-	-	-	2
CO3	K3	3	2	-	-	2	3	-	-	-	-	-	2
CO4	K3	3	2	-	2	2	3	-	-	-	-	-	2
CO5	K3	3	2	2	2	3	3	-	-	-	-	-	2
CO6	K3	3	2	-	-	1	3	-	-	-	-	-	2
Average		3.0	1.8	0.7	1.2	1.8	2.7	-	-	-	-	-	1.8

REFERENCEBOOKS

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION /REPRINT
1	Ramesh C. Chandan: Dairy-based Ingredients, Eagan Press	1997
2	De Sukumar: Outlines of Dairy Technology, 46th Edition, Oxford University Press	2019
3	Aneja, Mathur, Chandan & A. K. Bannerji: Technology of Indian Milk Products: DairyIndia Publication	2002

Course details

S. No.	Name of Experiment
1	Platform Tests of milk [COB, MBR Test, Alcohol Test, Sediment Test]
2	Determination of Fat content in Milk and Milk Products
3	Determination of SNF Content in Milk
4	Determination of Titratable Acidity in Milk
5	Determination of Overrun in Icecream
6	Preparation and Quality Testing of Butter
7	Preparation and Quality Testing of Butter oil / Ghee
8	Analysis of Adulteration in Milk and Milk products